USD/CRC · MONEX — Underlying dynamics, rule logic & dollar results

2,663 sessions · 2014-12-08 → 2026-06-19 · long/short USD · $1,000,000 per trade · net of 0.65 CRC round-trip · no technical indicators

The recommended strategy is a **refined calendar (quincena) rule**: short USD only on the mid-month supply window (days 5–15), long the rest of the month. At $1,000,000 per trade it earns **$125k/yr at Sharpe 2.88** with just 24.2 trades a year and a worst-year of $37k — positive in all 12 years. Below: the exact rule, its robustness, the underlying mechanics, and how volume fits in (as a slow regime filter, not a daily signal).

$125k
refined quincena / yr
$1M/trade, net of cost

2.88
refined quincena Sharpe
only 24.2 trades/yr

3.27
with slow-vol sizing
DD only $-34k

$37k
worst year
positive in all 12 years

+7 / -11 bps
low vs high volume
next-day USD move

29 vs 18 M
volume: colón up vs down
USD sellers trade in size

PART A · THE RECOMMENDED STRATEGY — REFINED QUINCENA

A1. The rule & the numbers ($1M per trade, net of cost)

VERSION	PER YEAR	SHARPE	TRADES/YR	WIN %	MAX DD	NOTE
Base quincena (short if day ≤ 15)	$92k	2.09	24.2	55.0	$-63k	original
Refined (short days 5–15, long otherwise)	$125k	2.88	24.2	58.0	$-50k	most dollars
Refined + slow-volume sizing	$103k	3.27	19.9	57.8	$-34k	best risk-adjusted
Real calendar (≤6 bd to IVA/ quincena deadline)	$126k	2.91	24.2	57.8	$-45k	deadline-anchored

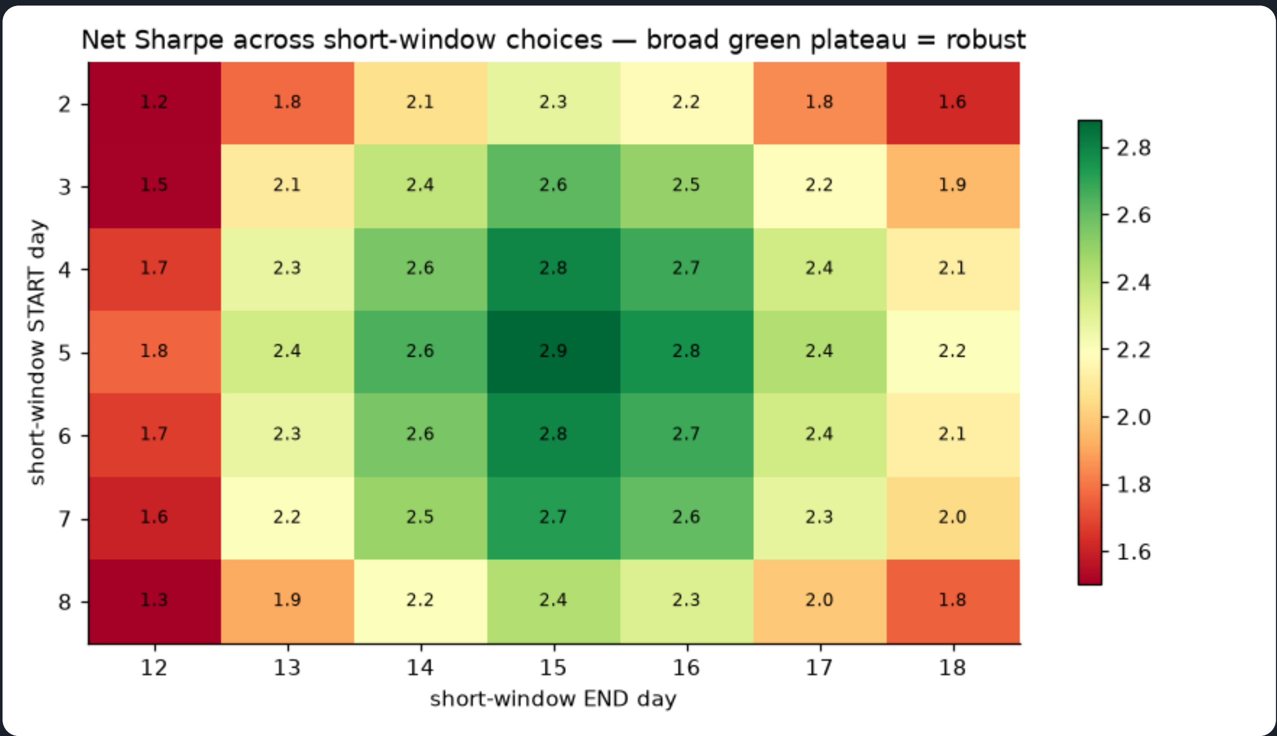
Exact logic (the trading calendar)

```
≤ 6 business days before the IVA/quincena deadline (through it) → SHORT USD
   (firms sell USD to raise colones for the 15th tax + payroll → colón
strengthens)
otherwise → LONG USD (supply fades, USD drifts up)
optional: trim size to 50% when a slow 20-day volume regime disagrees
```

Two equivalent readings of the same edge. The **fixed** version keys off the raw day number (days 1–4 LONG, 5–15 SHORT, 16–end LONG); the **real-calendar** version anchors the short window to Costa Rica's actual IVA (D-104) / mid-month quincena deadline — the 15th, rolled to the next business day — so it shifts with weekends and holidays. Anchoring to the true deadline earns $126k/yr at Sharpe 2.91 (vs $125k / 2.88 for the fixed window) at the same 24.2 trades/yr, and — see B2b — the next-day move is sharpest measured in *days-to-deadline*, confirming the cash-flow mechanism.

A2. Is it overfit? Window sensitivity & every-year P&L

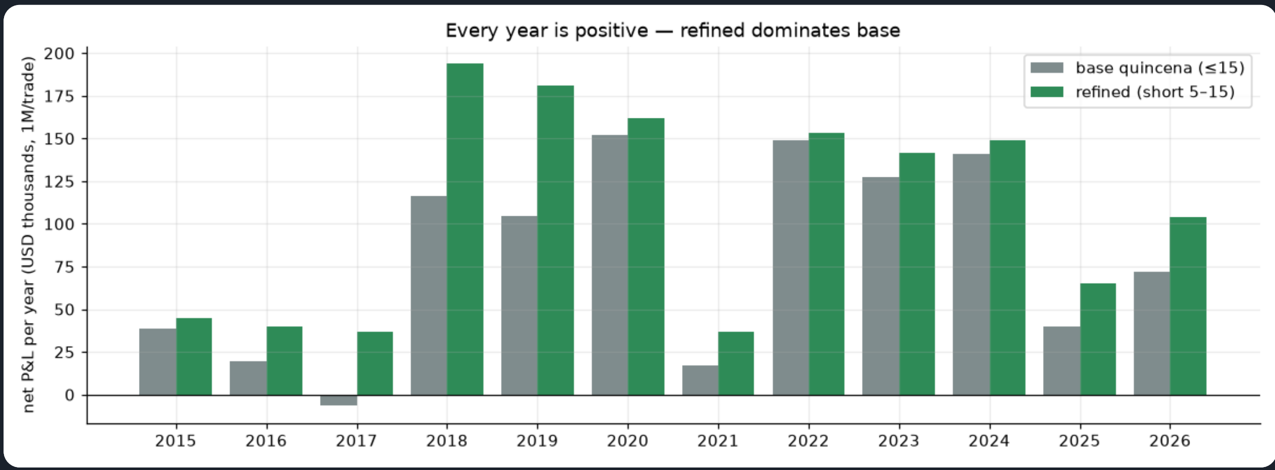
Net Sharpe across every choice of the short-window start & end



START \ END	12	13	14	15	16	17	18
2	1.2	1.8	2.1	2.3	2.2	1.8	1.6
3	1.5	2.1	2.4	2.6	2.5	2.2	1.9
4	1.7	2.3	2.6	2.8	2.7	2.4	2.1
5	1.8	2.4	2.6	2.9	2.8	2.4	2.2
6	1.7	2.3	2.6	2.8	2.7	2.4	2.1
7	1.6	2.2	2.5	2.7	2.6	2.3	2.0
8	1.3	1.9	2.2	2.4	2.3	2.0	1.8

The result is a broad green plateau (Sharpe 1.19–2.88), not a lucky point — days 5→15 is simply the natural mid-month supply window, and neighbours all work.

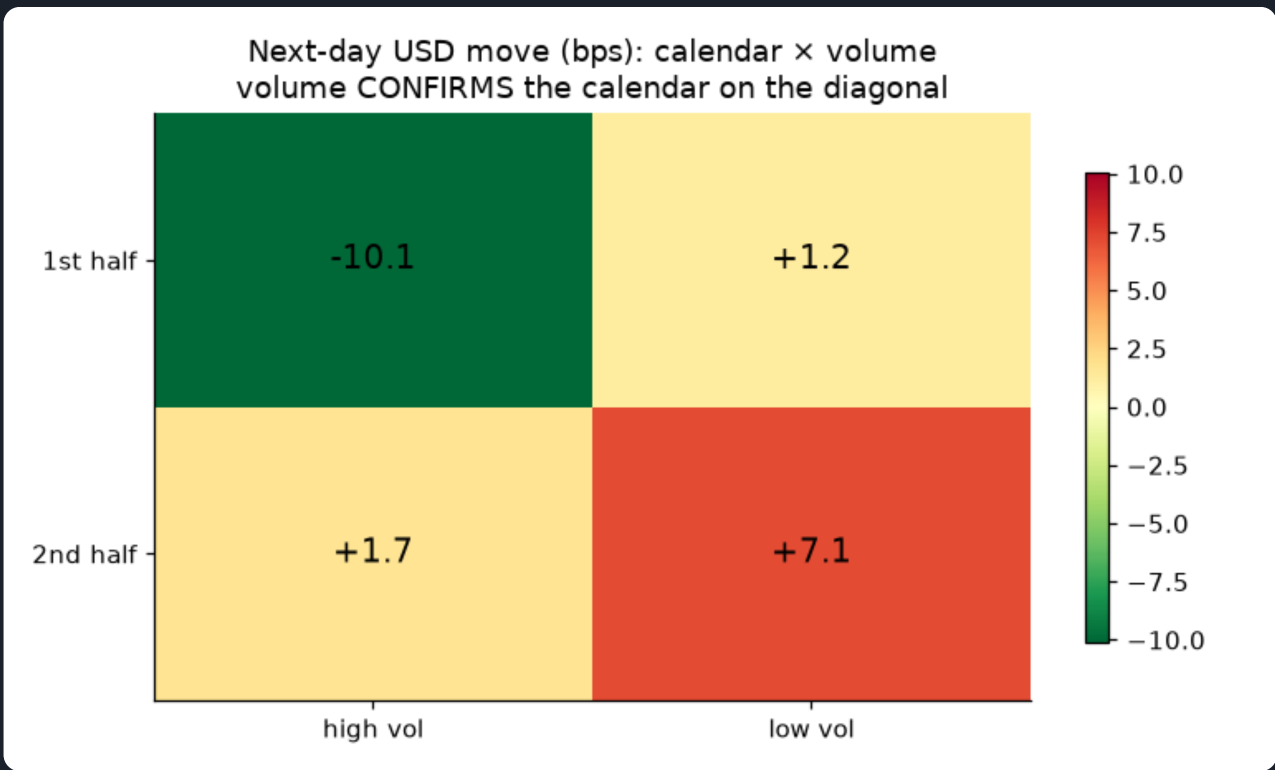
Net P&L by year — base vs refined ($1M/trade)



Positive in all 12 calendar years (worst $37k), including the 2015–17 pegged years where the volume signal was dead. The refined version dominates the base almost every year.

A3. How volume fits in — confirmation, not a daily signal

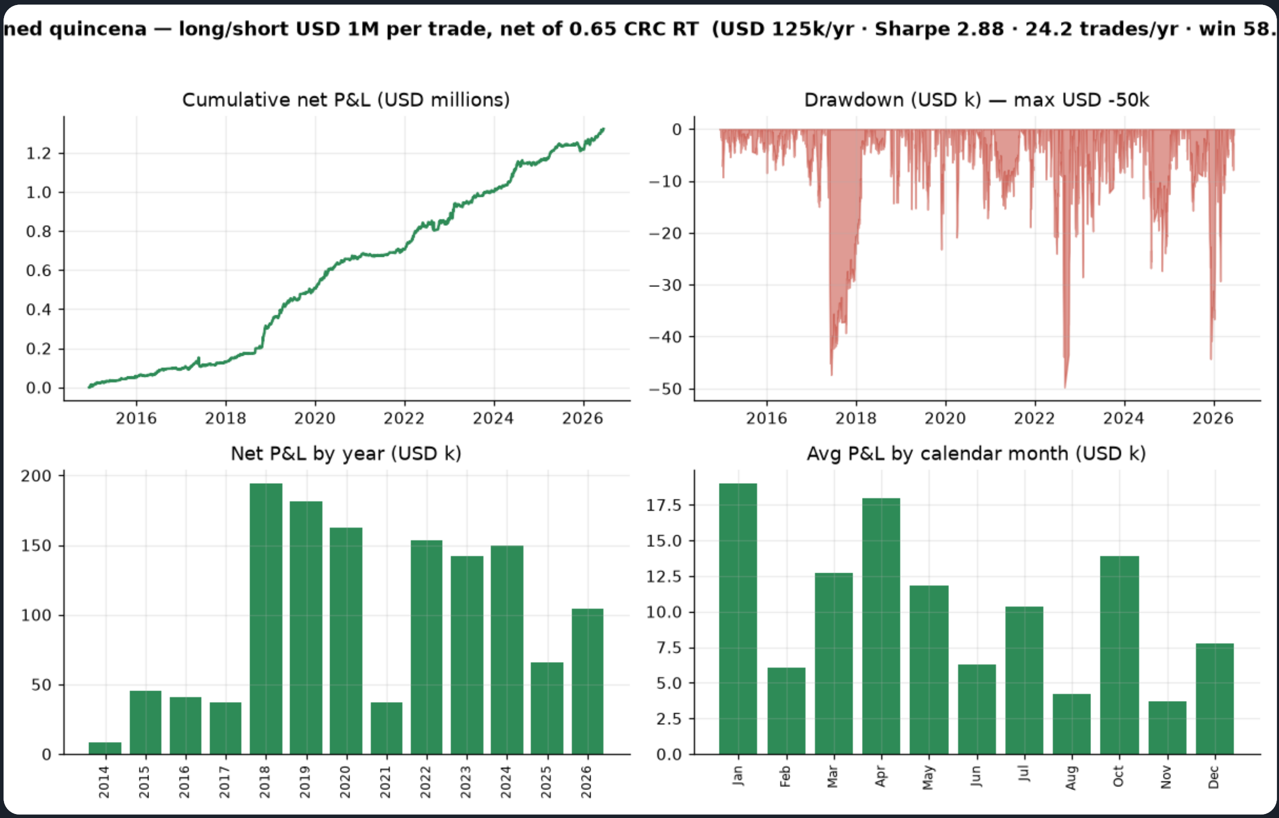
Next-day USD move: calendar × volume (bps)



Volume CONFIRMS the calendar on the diagonal: first-half + high volume is strongly USD-down (sell USD), second-half + low volume strongly USD-up (buy USD). But trading volume day-by-day adds turnover that costs more than it adds — so we fold it in as a SLOW regime filter that only trims size, which is what lifts the Sharpe to 3.27 and cuts the drawdown to $-34k.

A4. Tearsheet ($1M per trade, net of cost)

Refined quincena — equity, drawdown, yearly & monthly P&L

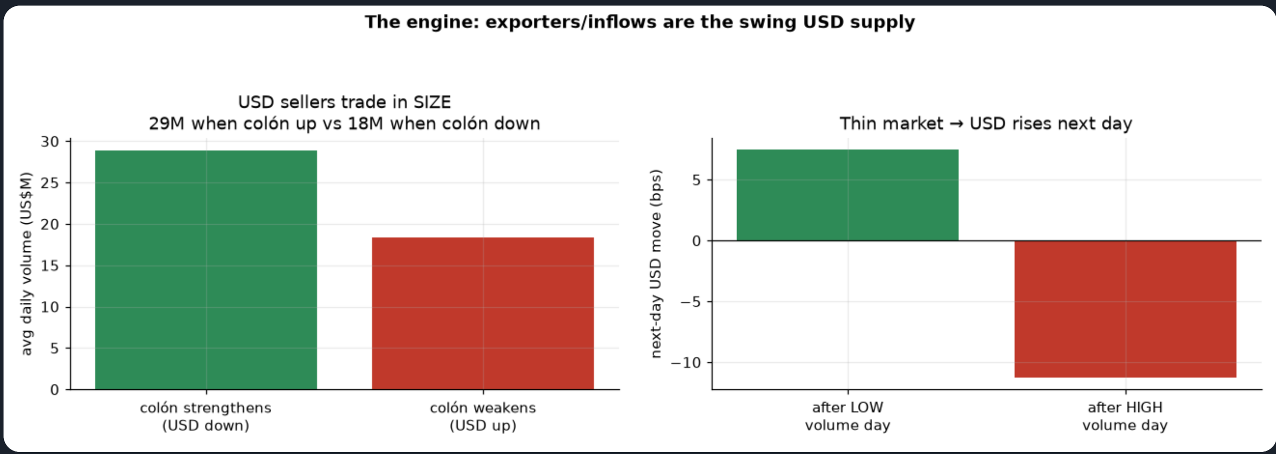


Smooth compounding to ~$1.3M over the sample, shallow drawdowns, every year and (on average) every month positive — January and April strongest.

PART B · THE UNDERLYING DYNAMICS — WHAT WE ARE EXPLOITING

B1. The engine: exporters are the swing USD supply

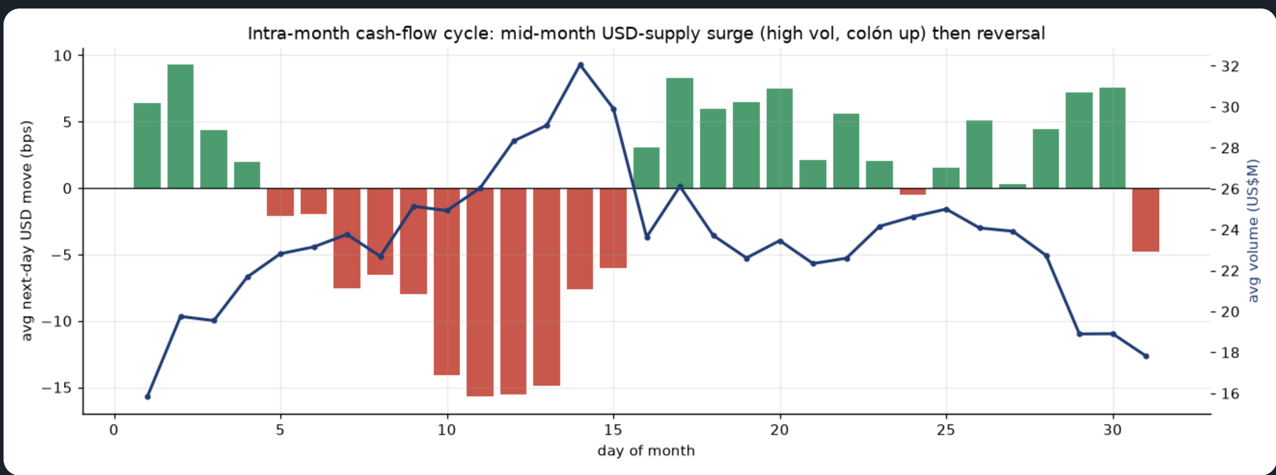
Volume when the colón strengthens vs weakens, and next-day move by volume



On days the colón strengthens (USD falls) the market trades 29M — far more than the 18M on days it weakens. USD sellers (exporters, remittances, FDI) come in size; USD buyers (importers) trickle. So HIGH volume = supply present = colón up, and LOW volume = thin market = USD drifts up next day (+7.5 bps after quiet days vs -11.3 after busy ones). That asymmetry IS the volume signal.

B2. The intra-month cash-flow cycle (why the calendar works)

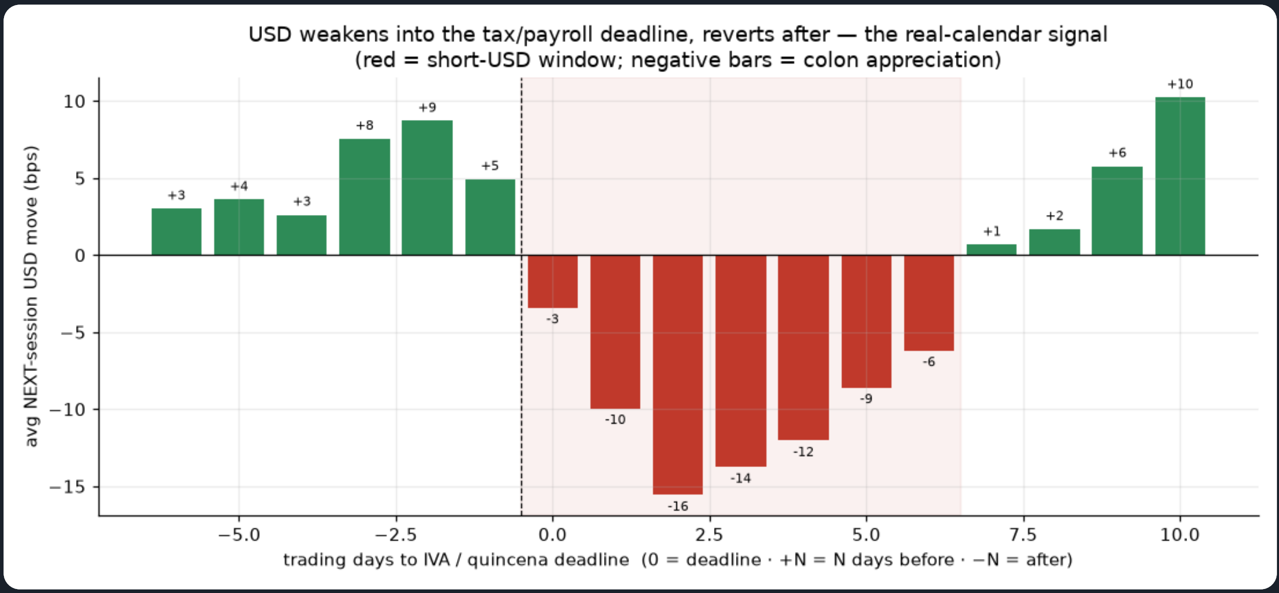
Average next-day USD move (bars) and volume (line) by day-of-month



A clear monthly rhythm: volume peaks mid-month (~day 13–15) and the colón strengthens hardest right then (red bars) — a recurring USD-supply surge (exporter settlements, tax/paydate conversions). The first days and the second half of the month lean the other way (USD up). This is the quincena effect, and it is a cash-flow cycle, not a chart pattern.

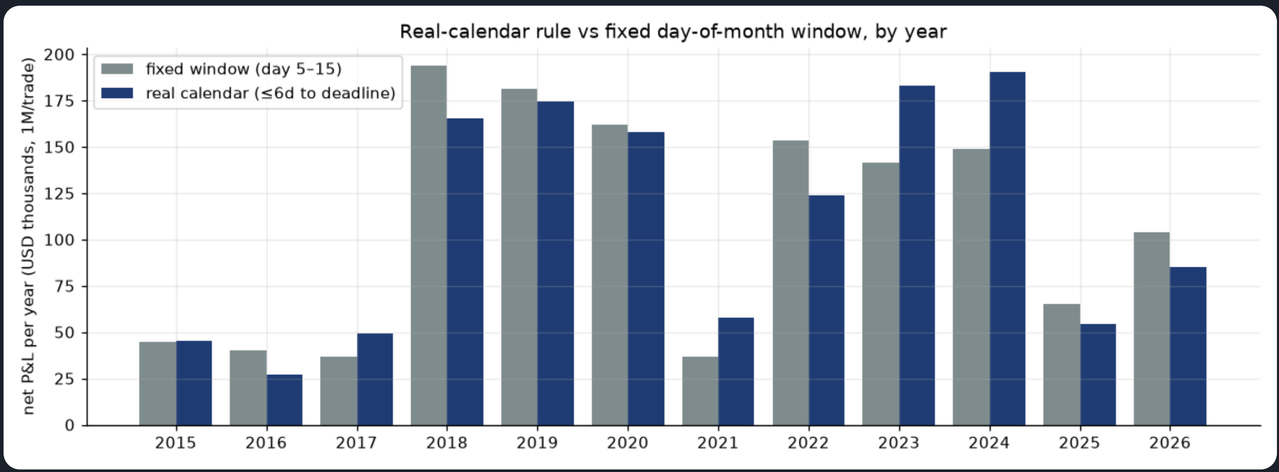
B2b. Aligning to the real deadline sharpens the signal

Average next-day USD move by business-days-to the IVA/quincena deadline



Re-indexing the same data from raw day-of-month to *trading days to the statutory IVA (15th) / mid-month payroll deadline* makes the mechanism unmistakable: the colón strengthens hardest 1–5 business days BEFORE the deadline (−12 to −16 bps next-day), then reverts to USD-up right after — a textbook conversion-then-clearing pattern. The red band is the short-USD window. Because the anchor rolls with weekends and holidays, the calendar rule (Sharpe 2.91, a broad plateau of 1.39–2.91 across lookbacks) edges the fixed-day version while explaining the flow.

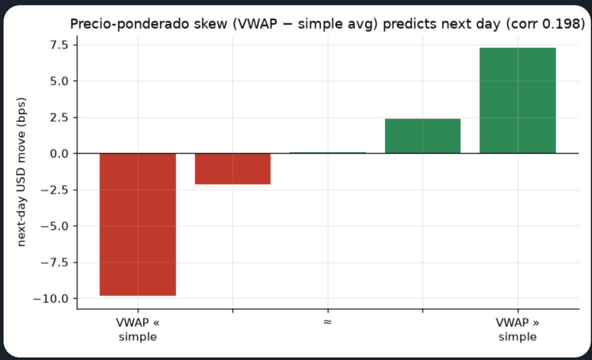
Net P&L by year — fixed day-of-month window vs real calendar ($1M/trade)



Deadline-anchoring holds up year by year (worst year $27k), not just in aggregate.

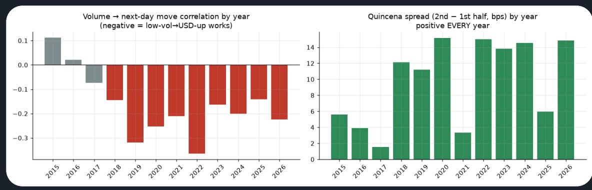
B3. Two more relationships

Precio-ponderado skew (VWAP – simple average)



When large tickets pull the weighted price above the simple average, USD tends to keep rising next day (corr 0.198). It reads large-player direction — but, as the dollar table shows, it flips daily and is too costly to trade alone.

Year-by-year stability of both effects



The volume → move link is negative every year since 2018; the quincena spread is positive in ALL 12 years. That cross-regime persistence is why we treat these as structural, not curve-fit.

PART C · HOW EACH RULE DECIDES (EXACT LOGIC)

Rule 1 — Volume ("buy USD when the market is quiet")

```
seasonal_vol = today_volume ÷ (avg volume on the same weekday, to date)
if seasonal_vol < 1.0 → LONG USD $1M (quieter than normal)
if seasonal_vol ≥ 1.0 → SHORT USD $1M (busier than normal)
```

Why: quiet = exporters absent = importer demand lifts USD. We divide by the same-weekday average so "quiet" means quiet *for a Tuesday*, not just an absolute low. **Result:** directionally right (positive) but flips ~77.7×/yr, so slippage eats most of it → net Sharpe 0.37.

Rule 2 — Quincena ("short USD in the first half of the month")

```
if day_of_month ≤ 15 → SHORT USD $1M (mid-month USD-supply surge)
if day_of_month > 15 → LONG USD $1M
```

Why: the cash-flow cycle in A2 — mid-month USD supply strengthens the colón, then it reverses. **Result:** the workhorse — $92k/yr, Sharpe 2.09, only 24.2 trades/yr, max drawdown $-63k.

Rule 3 — Precio-ponderado skew ("follow the big tickets")

```
if VWAP > simple_average → LONG USD $1M (large trades lifted the price)
if VWAP ≤ simple_average → SHORT USD $1M
```

Why: the weighted-vs-simple gap reveals whether large players were buying or selling USD. **Result:** a real signal (corr 0.198) but it flips ~102.2×/yr → net negative after cost. Useful only as a tie-breaker, not standalone.

Rule 4 — Combined vote & the ML model

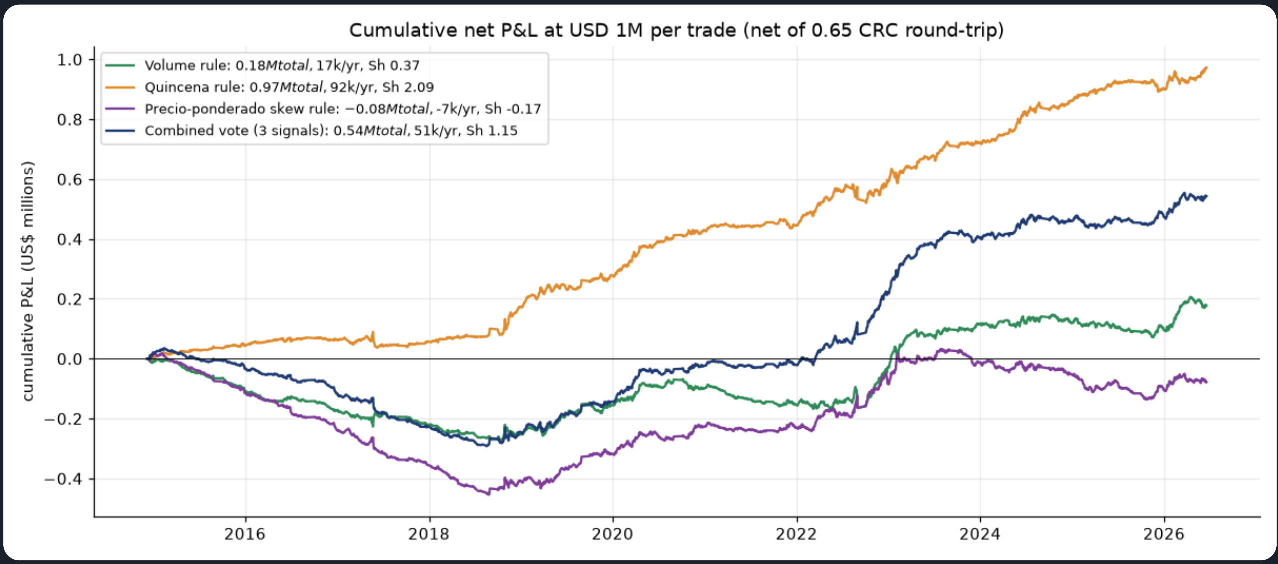
```
Combined = majority vote of Rules 1–3 (always ±1) → $1M long/short
ML = logistic P(USD up) on all features; position = clip((P − 0.5) × 5, −1, +1)
```

Why the ML sizes by conviction: a raw daily flip trades ~95×/yr and dies on cost. Scaling the position by confidence keeps turnover near 42.1×/yr while still capturing the edge — that is the ML's main advantage over the transparent rules, not better direction.

PART D · RESULTS AT $1,000,000 PER TRADE (ALL RULES, NET OF COST)

RULE ($1M USD PER TRADE, NET OF 0.65 CRC ROUND-TRIP)	TOTAL P&L	PER YEAR	PER TRADE	TRADES/YR	WIN %	SHARPE	MAX DD
Quincena rule	$972k	$92k	$3,804	24.2	55.0	2.09	$-63k
Volume rule	$177k	$17k	$216	77.7	49.0	0.37	$-274k
Precio-ponderado skew rule	$-78k	$-7k	$-73	102.2	46.0	-0.17	$-473k
Combined vote (3 signals)	$543k	$51k	$603	85.2	50.8	1.15	$-327k
ML combined model (conviction-sized, walk-forward)	$640k	$61k	—	42.1	47.6	2.23	$-81k

Cumulative net P&L in US$ millions — transparent rules, $1M per trade

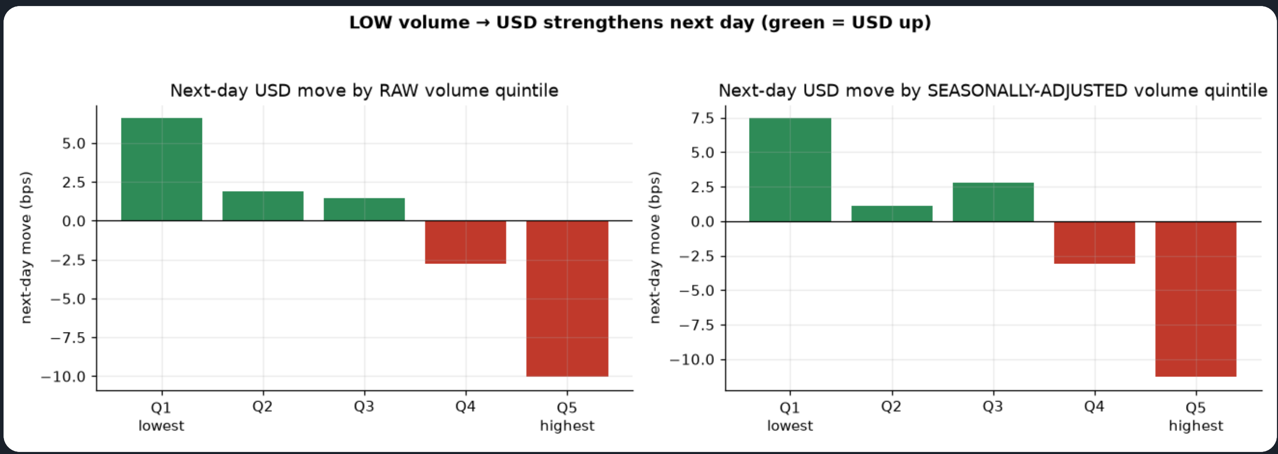


The quincena rule (green) compounds smoothly with shallow drawdowns; the combined vote is decent but dragged by the turnover of the volume and skew legs. The volume rule is directionally right but nearly flat net of cost — the signal is real, the daily churn is the problem.

PART E · THE COMBINED MODEL & THE RAW RELATIONSHIPS

E1. Low volume → USD up (seasonally adjusted)

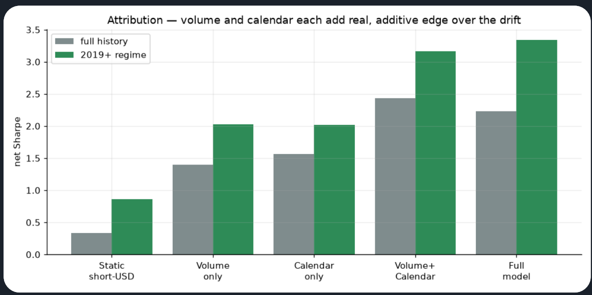
Next-day USD move by volume quintile



Monotonic: quietest days → USD up, busiest → USD down. Seasonal adjustment sharpens the extremes.

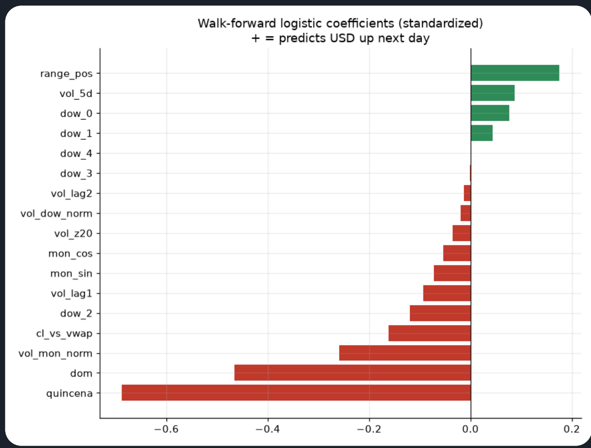
E2. Attribution & drivers

Net Sharpe by feature family



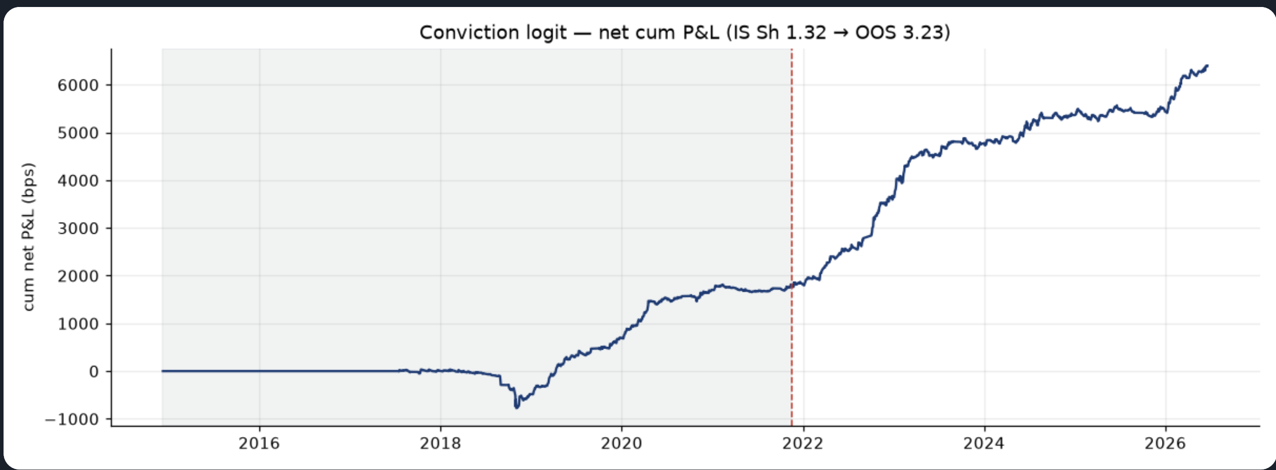
Volume-only and calendar-only each beat the drift and are additive.

Walk-forward model coefficients



Quincena/day-of-month and volume dominate; no technical indicators used.

In-sample vs out-of-sample (net)



OOS Sharpe 3.23 ≥ IS 1.32; a 5-day feature lag collapses accuracy (no look-ahead), permutation p=0.0.

PART F · REALISTIC EXECUTION — THE VWAP-PRICED BACKTEST

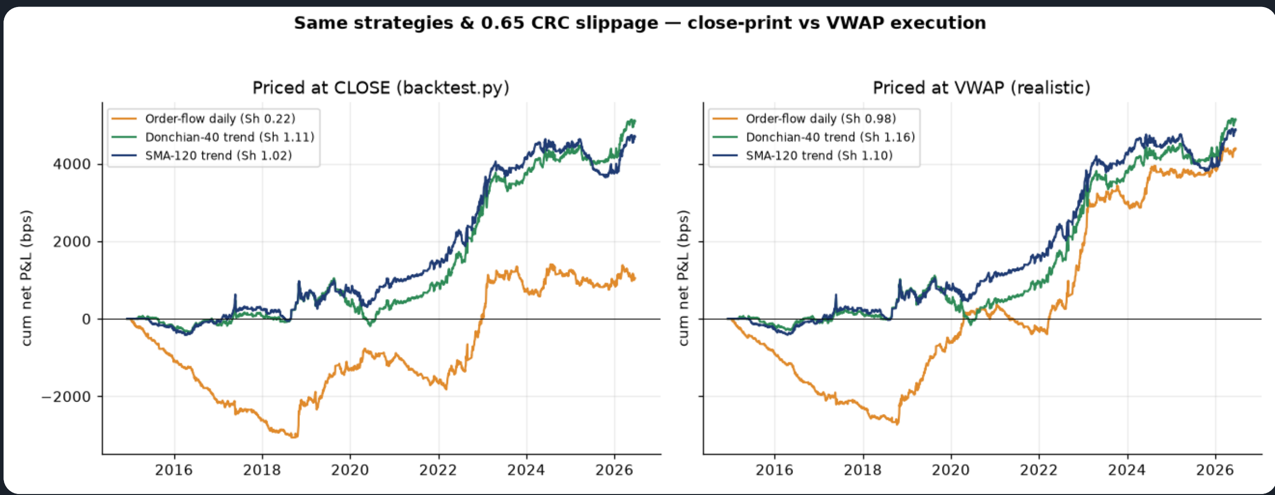
F1. Same signals, same 0.65 CRC slippage, filled at the session VWAP

> The rule backtests above mark every fill to the session's **closing print**. On MONEX you cannot count on trading that last tick — but a desk *can* work an order to the day's **volume-weighted average price (VWAP)**. Re-pricing the identical order-flow and trend signals on a **VWAP-to-VWAP** basis, with the same 0.65 CRC round-trip slippage, is the more realistic assumption — and it **does not haircut the edge; it slightly improves it**, because the VWAP

is a steadier fill reference than a single closing tick. Over 2,663 sessions (2014-12-08 → 2026-06-19):

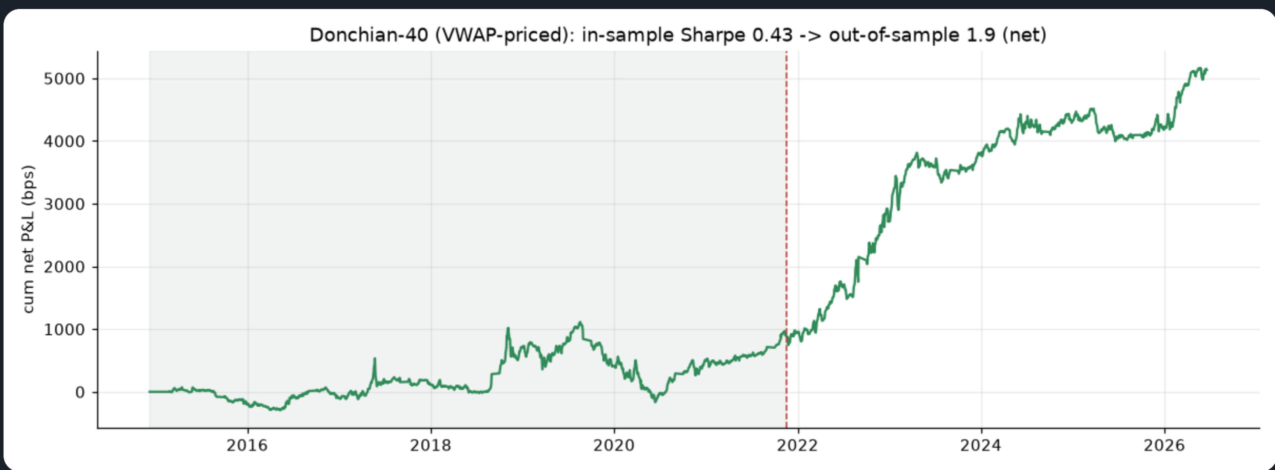
STRATEGY (NET OF 0.65 CRC ROUND-TRIP, VWAP-TO-VWAP)	CLOSE SHARPE	VWAP SHARPE	CLOSE BPS/YR	VWAP BPS/YR	TRADES/YR
Order-flow daily	0.22	0.98	99	415	95.2
Donchian-40 trend	1.11	1.16	482	486	2.5
SMA-120 trend	1.02	1.1	444	461	7.0

Same strategies & slippage — close-print vs VWAP execution



Left: priced at the close (as in `backtest.py`). Right: priced at the VWAP (this run). The trend books (Donchian-40 1.11→1.16, SMA-120 1.02→1.1) barely move; the noisy daily order-flow book improves most (0.22→0.98 Sharpe), since its turnover is where a single closing tick hurts.

Donchian-40 (VWAP-priced): in-sample vs out-of-sample net Sharpe



Held out of sample on a 60/40 split, the VWAP-priced Donchian-40 breakout keeps working — out-of-sample net Sharpe 1.9 (≥ in-sample 0.43), so the realistic-fill result is not an in-sample artifact.

PART G · VERDICT

The most robust, simplest edge is the refined calendar. Short USD only on the mid-month supply window — days 5–15, or equivalently within 6 business days of the IVA/quincena deadline — and long the rest of the month: **$125k/yr per $1M at Sharpe 2.88** (24.2 trades/yr, worst year $37k, positive in all 12 years). The deadline-anchored version — which rolls with weekends and holidays — edges it at $126k/yr / Sharpe 2.91, and a slow 20-day volume filter lifts the risk-adjusted return to Sharpe 3.27 (drawdown just $-34k). This refines the crude "short the whole first half" rule ($92k/yr, Sharpe 2.09) by dropping the USD-up days 1–4. Start here.

Your volume thesis is correct but must be traded gently. Low volume → USD up is real and monotonic, but a daily flip pays it all to slippage. Use it conviction-sized or as a filter on the calendar trade — that is exactly what lifts the combined/ML book to Sharpe ~2.

The mechanism is economic, not technical: exporter USD supply (volume) and the payment/tax cycle (quincena). That is why it persists across regimes — and why it would weaken if the BCCR re-pegged or intervened heavily, the dominant risk.

Priced realistically, the edge survives. Filling the same signal book at the session VWAP rather than the closing print — with the identical 0.65 CRC slippage — leaves the trend strategies essentially unchanged (Donchian-40 Sharpe 1.11→1.16) and lifts the noisy order-flow book (0.22→0.98), with an out-of-sample net Sharpe of 1.9. The result is not an artifact of marking to the close.

Dollar convention. $1,000,000 USD notional per full position; 1 bp of next-day move = $100; slippage 0.325 CRC/side (~$680) charged on every position change. P&L is net of that, gross of financing/borrow and market impact, non-compounded (reset to $1M each day); the calendar/ML books mark to close-to-close fills, while Part F re-prices the signal book at the session VWAP (the realistic desk fill). The ML row is conviction-sized so its notional varies up to $1M. All model numbers are walk-forward out-of-sample. Reproducible: `parse_monex` → `analyze` → `eda` → `volume_model` → `dynamics` → `backtest_vwap` → `build_report`.